Inverse Heat Modelling and Scientific Diagnostics

AI/ML Engineer Intern take-home — report

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Data status (read first). Only the assignment PDF was received; the versioned data bundle and the Stage-4 diagnosis dataset were not attached (requested by email). Every result below is therefore on **synthetic** data generated to the schema the brief describes, with ground truth stored separately and read only for scoring; such numbers carry the mark [SYN]. The pipeline takes any bundle in the documented format (README), so a real bundle needs a one-line config change.

1 Formulation

Model. $u_t = \partial_x[a(x)u_x]$, $a = a_0 e^{\beta(2x-1)}$, $u(0,t) = g(t)$, $-a(1)u_x(1,t) = h[u(1,t) - T_a(t)]$, readings $y = u(x_s,t) + b_s + \varepsilon$, $b_{\text{ref}} = 0$. Physically: Fourier conduction with a graded conductivity (a_0 = mid-rod value, $e^{2\beta}$ = right/left ratio), a clamped heater at x=0 and Newton cooling at x=1. With $a \approx 0.07$ heat travels $\sqrt{at} \approx 0.26$ in the experiment, so the Robin end is seen mainly by the right-most sensors: the a-priori reason h is the hardest parameter. Heat leaving at x=1 crosses conduction and film resistances in series, so a_0, h and offsets can trade off.

Inverse problem. $\theta = (a_0, \beta, h, b_s)$, all box-constrained to the brief's ranges; minimise $\frac{1}{2}\sum_k [(u_k(\theta) + b_{s(k)} - y_k)/\sigma_k]^2$ (Gaussian likelihood with the supplied noise sd). Offsets are *persistent* per sensor across experiments, enter linearly (exact Jacobian columns), and the reference sensor anchors the level (without it offsets and smooth physics changes exchange misfit). Uncertainty: Gauss–Newton/Laplace covariance $(J^\top J)^{-1}$ — local, linearised, conditional on the model — with guards (§5) that widen or withhold it. **Assumptions:** independent Gaussian noise with known sd (tested: residual autocorrelation), model form correct (tested: nested-model and fault tests), piecewise-linear inputs. **Data audit.** Rows are cleaned explicitly and logged with their raw CSV line: in the public bundle[SYN] 10 `bad`-flagged, 8 conflicting repeated-ID, 6 exact duplicate, 5 missing and 2 out-of-domain rows were dropped; 6 `suspect` rows kept and logged. Identity key =(experiment, obs_id); conflicting copies are all dropped (no silent tie-break). Split: fit E1–E3, validate E4 (labels never reach a fitting routine — tested by corrupting them), predict E5 from inputs only.

2 Stage 1 — reference solver and inference

Cell-centred finite volumes with exact face diffusivities (conservative: energy changes only through boundary fluxes), the Robin end as half-cell conduction in series with h, Crank–Nicolson with Rannacher start-up (four backward-Euler quarter steps; avoids CN ringing when $f(0) \neq g(0)$). **Independent verification:** (i) manufactured solution with a hand-derived source — observed order 1.993–1.999 under joint space–time refinement (Fig. 1a), RMSE 3.3e-05 at the production grid (100 cells, 400 steps) vs noise $\approx 10^{-2}$; (ii) steady state against the closed form, error 2.9e-06; (iii) discrete energy balance to 2e-16; (iv) maximum principle with a discontinuous corner. Synthetic data are generated on a 4× finer grid than inference (no "inverse crime"). Fitting: bounded trust-region least squares, multistart (Latin hypercube), central differences for (a_0, β, h). Known-parameter recovery is a test on separate synthetic data (all three within 4σ, offsets within 0.01); coverage over 20 datasets is in §4.

Result (Table 1). The variable model recovers all three parameters inside their intervals with χ^2_ν=1.12. The constant baseline is the cautionary case: its error against noisy readings is close to the primary (0.0139 vs 0.0131) yet its latent error is 0.0084 vs 0.0012 and its latent intervals cover the truth only 0.33 of the time — offsets absorb the missing gradient. It is rejected automatically by the nested test ($\Delta\chi^2$=150, §5).

3 Stage 2 — inverse PINN

Design. One 4×48 tanh MLP per experiment (inputs scaled to $[-1, 1]$); shared (a_0, β, h) bounded by sigmoids, offsets by 0.03 tanh. Losses in noise units (divided by the median sensor variance) so a constraint violated by one noise level costs about one misfitted reading: L_{obs} (whitened), L_{pde} with the *conservative* operator by autograd (flux $= a(x)u_x$, then ∂_xflux — the $a'(x)u_x$ term appears automatically; unit-tested against the hand-derived MMS source to 10^{-10}), L_{ic}, L_{bc} (Dirichlet + Robin). Uniform collocation (2000/experiment) resampled every 250 Adam steps, then L-BFGS on a *fixed* set (resampling breaks its line search). Adam lr decays 10×; physics frozen for 1500

Table 1: Public bundle^{SYN}: estimates (95% Laplace intervals) and errors. Truth a_0=0.104, β= − 0.361, h=0.322. "noisy" = vs readings y; "latent" = vs noise-free u at E5 queries. PINN: mean over 3 seeds, [min,max] across seeds (a spread, *not* an interval).

	FV, variable $a(x)$ (primary)	FV, constant a (baseline)	inverse PINN
a_0	0.107 [0.100, 0.113]	0.120 [0.109, 0.131]	0.094 [0.091, 0.096]
β	-0.405 [-0.471, -0.339]	0 (fixed)	-0.426 [-0.444, -0.413]
h	0.330 [0.272, 0.388]	0.278 [0.215, 0.340]	0.174 [0.158, 0.185]
χ^2_ν (train)	1.12	1.48	—
train RMSE (noisy)	0.0135	0.0158	0.0137
val E4 RMSE (noisy) / 95% cover.	0.0124 / 0.96	0.0142 / 0.92	0.0164 / —
test E5 RMSE noisy / latent^{SYN}	0.0131 / 0.0012	0.0139 / 0.0084	0.0169 / 0.0144
test latent-interval coverage^{SYN}	0.99	0.33	—
fit time (laptop CPU)	6.1 s	5.6 s	243 s/seed + 59 s adapt.
energy-balance defect	5e-16 (discrete)	—	0.022 (relative)
status (automatic)	ok	weak_evidence	3/3 weak_evidence

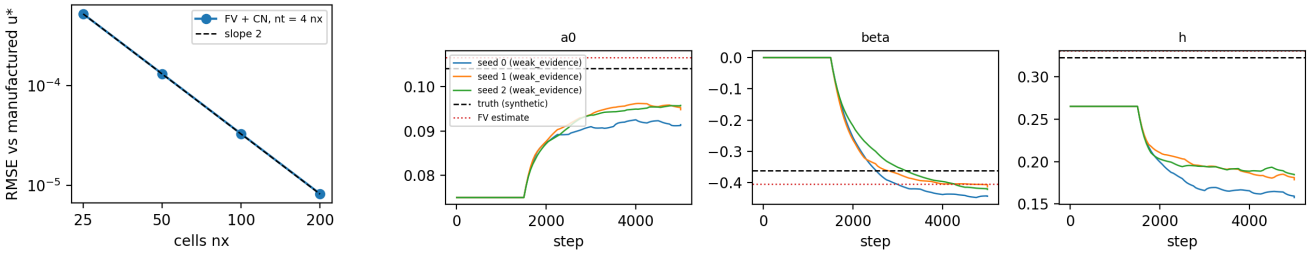


Figure 1: (a) MMS refinement, slope 2. (b) PINN parameter trajectories, public bundle^{SYN}, 3 seeds: β settles near the FV value (dotted); a_0 ends 10% low and h far below truth (dashed).

warm-up steps. Unseen experiments: physics-only adaptation (fresh field net, PDE+IC+BC, parameters frozen; timed: 59 s/seed).

Development (dev bundle, one change per trial; `docs/evidence`). (1) no warm-up: h collapsed onto its bound — an untrained field makes the Robin term cheapest at $h \to 0$ and the sigmoid saturates; (2) warm-up: a_0, β correct, h 3× low with the right-most offset absorbing it; (3) w_{bc}=10: hypothesis "BC weight pins h" **refuted** (h worse); (4) 2× steps: marginal; (5) w_{pde}=10: worse (stiffer). Mechanism: soft constraints tolerate ∼1 noise level of physics error, enough for the field to bend between the last sensor (x=0.85) and the Robin end so a wrong h satisfies the BC. Tuning stopped there.

Result. On the public bundle^{SYN} the PINN reproduces the readings (L_{obs}=1.16, train RMSE 0.0137) with small seed spread, yet its parameters are off: h by 46% (4.9–5.8 FV standard deviations from the FV estimate) and a_0 by 10% (3.1–4.4 FV sd); only β agrees with FV (within 1.2 FV sd; truth errors 18% PINN, 12% FV). Seed spread is *not* uncertainty. Agreement of the PINN field with an FV field at the PINN's own parameters (RMSE 0.0029) shows consistency only. The accuracy question is answered by an exact-physics likelihood-ratio check (§5): all seeds lie outside the FV fit's joint 99.9% region ($\Delta\chi^2$ 37–58 vs 26.1) and are reported `weak_evidence`. **Conclusion:** the PINN does not outperform the numerical baseline here; the FV fit is primary.

4 Stage 3 — investigation

Uncertainty (Fig. 2b). 95% Laplace intervals for a_0, β, h on 20 independent synthetic datasets: coverage 20/20, 20/20, 20/20 (Wilson lower bound 0.84; n is small — a sanity check, not calibration). Data uncertainty isolated by redrawing only the noise 20 times: empirical sd / Laplace sd = 1.25 (a_0), 1.24 (β), 1.29 (h), about 1.6–1.8 times the ±16% sampling error of an sd from 20 draws (hinting mild optimism), whereas across datasets the standardised errors have sd 0.74–0.77 (hinting mild conservatism); at n=20 neither is conclusive and coverage is nominal. Seed variability is a separate quantity: zero for the deterministic FV fit (multistarts agree), the PINN spread in Table 1 for the PINN. Dependence assumption tested with AR(1) noise (ρ=0.6): plain Laplace intervals cover 18, 18, 15 of 20; the autocorrelation guard flags 20/20 such datasets and widens intervals by $(1+\hat{\rho})/(1-\hat{\rho})$, giving 19, 20, 20 of 20; it flags 0/20 independent-noise datasets.

Controlled experiments (Fig. 2a; one change each, fixed evaluation = latent RMSE on held-out E4). Truth $(0.07, 0.45, 0.25)$. *Coverage:* removing sensors at $x \geq 0.7$ widens the h interval 5.3×; early times only ($t \leq 0.3$) widen it 5.3× and β's too; 4× sparser sampling widens all intervals but estimates stay within them; dropping left sensors mainly widens β (1.5×). *Assumptions:* constant a gives 9× the held-out latent error; offsets forced to zero push β to 0.700. Hypotheses were stated before running (code, `ABLATION_HYPOTHESES`).

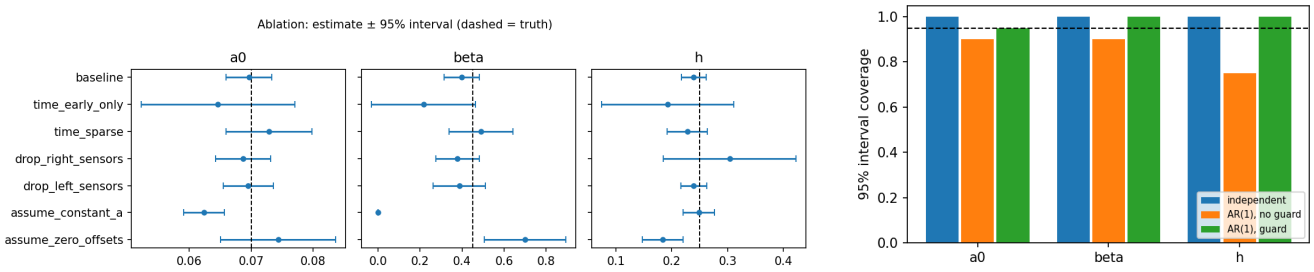


Figure 2: (a) Ablation: estimates ±95% (dashed: truth). (b) Interval coverage: independent noise; AR(1) noise without and with the guard.

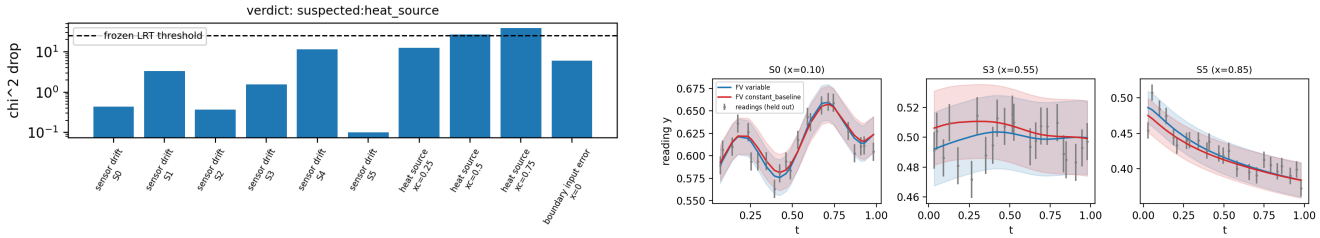


Figure 3: (a) Sealed blind test: $\Delta\chi^2$ per hypothesis vs frozen threshold. (b) Held-out E4^{SYN}: readings vs FV variable (blue) and constant (red) predictions with 95% reading intervals.

Experimental design. One new sensor at x=0.95 (24 readings/experiment, unknown offset) vs the same 24 readings spread over existing sensors. Rule, fixed before truth: choose the design with the smaller linearised posterior sd of h at a *pilot* fit (Fisher information incl. the new offset). The ambiguity addressed is the h–conduction–offset trade-off at the Robin end. The rule chose the new sensor in 10/10 datasets; it narrowed the h interval in 10/10 (mean width 0.0376 vs 0.0487; base 0.0531) but mean $|h|$ error did not measurably differ at n=10 (0.0078 vs 0.0072) — a precision gain, not a demonstrated accuracy gain.

The three questions. *Can sensor bias mask incorrect physics despite low temperature error?* Yes, more so as the true gradient weakens. With β=0.45 (ablation^{SYN}) the constant-a model with free offsets fits the training readings at χ^2_ν=1.17 (limit 1.20: *accepted*) while its held-out latent error is 9× larger. With β=0.6 (5 datasets^{SYN}) offsets still absorb most of the misfit (train RMSE 0.0150 vs 0.0105) but χ^2_ν=2.09 exposes it. Detection needs model comparison or held-out prediction, not training fit alone. *Which parameters become weakly identifiable?* h first (right-end coverage or late times), then β (early times); a_0 stays well determined. *Which additional measurement helps, and why?* A sensor near the Robin end: it observes the boundary gradient directly, breaking the h/conduction/offset trade-off — but it brings its own unknown offset, which the selection rule accounts for.

5 Stage 4 — reliability, diagnosis, recovery

Automatic status. Every result carries `ok`/`weak_evidence`/`failed` from rules fixed in code, none using truth: χ^2_ν above $1+3\sqrt{2/\nu}$; positive residual autocorrelation (z>3); held-out predictive χ^2; nested LRT constant-vs-variable a (p=10^{-3}); interval >50% of the admissible range ⇒ estimate *withheld*; estimate on a bound; multistart disagreement; PINN exact-physics recheck; fault-hypothesis scan. Most were added in response to a measured miss; false-alarm rates were measured on correct models.

Model diagnosis. Rules frozen in `configs/diagnosis_rules.yaml`. Observations (residual trends, χ^2, early/late parameter shift) are reported separately from causal hypotheses, each a nested one-parameter model: drift of sensor s (y+rt, every sensor), Gaussian heat source (centre scanned), constant bias of the recorded g. Evidence = $\Delta\chi^2$ (LRT), with the extra parameter's estimate ± sd as sensitivity. Threshold v2 = 15.1 (Bonferroni over 10 hypotheses, family p=10^{-3}; it replaced an ad-hoc 25, which would have missed 2 of the sweep's detections); over 13 fault-free bundles the largest $\Delta\chi^2$ is 6.8 (0 false alarms). Validation^{SYN}: 12/12 correct. Fault-size sweep^{SYN}: 13/22 detected, 1 ambiguous, 0 of 13 named verdicts wrong. Missed faults reach drift rate 0.01/unit time, source Q=0.03 and boundary bias 0.02; undetected, they still bias β by up to 0.09. **Blind test:** rules committed → sealed bundle (fault drawn from OS randomness, truth hashed) committed → verdict committed → unsealed. Verdict: *suspected:heat source* ($\Delta\chi^2$=38, Q=0.051 ± 0.009 at the nearest scanned centre); truth: heat source, Q=0.082 at x=0.78 (width differs; integrated heat agrees within 6%). The base fit there reported `ok` with h=0.178 (true 0.288; 0.279 once the source is modelled) — which is why the fault scan now runs inside `evaluate` (public bundle: no fault detected).

Recovery and visibility. Checkpoints carry a fingerprint (config + data hash); writes are atomic. A run killed mid-training and resumed reproduces the uninterrupted parameters exactly (max difference 0); a checkpoint from

another config is refused; a truncated newest file is skipped; an injected non-finite loss is logged, rolled back and the lr halved (all pass). Every result JSON logs data sha256, config + hash, seed, git commit/dirty flag, hardware, threads, versions, wall time.

One misleading result, investigated. In the first public evaluation a PINN seed passed the recheck with h=0.185 (true 0.322): χ^2_ν=1.2077 against a limit of 1.2078. Investigation: an absolute χ^2_ν test has little power because a wrong h moves χ^2_ν by only 0.09 above the optimum. Fix: test the PINN parameters against the FV optimum on the same data (LRT, χ^2_k); it rejects this case by a wide margin and is pinned by a regression test. Testing also found six bugs (IDs restarting per experiment were all dropped as conflicts; a NaN input passed silently; unknown flags were trusted; per-experiment query IDs were rejected; a garbage numeric cell crashed the loader; the smoke run failed on a fresh clone), all fixed.

Compute. Laptop i7-1165G7 (4 cores), 16 GB, no GPU, 4 torch threads. FV fit 6.1 s; PINN 243 s/seed + 59 s physics-only adaptation; full public evaluation 16 min; CPU smoke run of every stage 3.4 min (PASS; budget 10 min).

6 Claims, limitations, time

Supported^{syn}: FV recovery with calibrated intervals under the stated noise model; h is the weakly identified parameter and a Robin-end sensor improves its precision; guards detect dependent noise, wrong diffusivity form, PINN parameter inconsistency and three fault types above measured sizes. **Not supported / unresolved:** anything on real data; faults below the detection limits bias β silently; Laplace intervals are local; the PINN's h; design accuracy gain (only precision shown); coverage from 20 datasets. **Distinctions kept:** training fit (Table 1 row 5) vs held-out prediction (rows 6–7) vs parameter recovery (synthetic only) vs physical consistency (energy balance, PINN–FV agreement). **Time spent:** build and experiments ≈5 h (24 Oct 21:50 – 25 Oct 02:30 IST, AI-assisted); review and walkthrough preparation on 25 Oct (author).

Appendix — AI assistance and verification (1 page)

How AI was used. Claude (Anthropic) was used as a pair-programmer throughout, as the brief permits: drafting code, running experiments, and drafting this report. I directed the plan, decisions and trade-offs; the AI proposed and implemented them; decisions such as dropping PINN tuning, adding guards and tightening thresholds were discussed and approved explicitly.

How the output was verified — by evidence, not trust.

- *Independent references:* the solver is checked against a manufactured solution (hand-derived source), a closed-form steady state, discrete energy conservation and the maximum principle; the PINN operator is checked against the same hand-derived source.
- *Known truth:* recovery, coverage, ablation, design and diagnosis studies use synthetic data with truth stored separately and read only for scoring; a monkeypatched file guard proves fit/predict never open it; corrupting validation labels leaves training estimates bit-identical.
- *Pre-registration:* methods and diagnosis rules were committed before the final evaluation and the sealed blind test; the git history shows the order. Rule changes (one-sided autocorrelation test, Bonferroni threshold) are declared with the evidence that motivated them.
- *Adversarial testing:* 58 automated tests (black-box: metamorphic shift/scale invariance, renamed IDs/layouts, row order, edge-of-box truths, malformed inputs, CLI; white-box: autograd operator, Jacobian, decision branches, interval algebra). They found six real bugs (one by fuzzing the loader), one wrong test premise and, on a fresh clone, a dependency-pin gap (installation now uses the full lock file).
- *AI errors caught:* an ad-hoc threshold, a guard with too little power, a premature claim that a status rule was protective (it let a 9× error through), a mis-stated clock time. Each was challenged, re-measured and corrected; the over-engineering review removed dead code and test hooks from production paths.
- *No hand-typed numbers:* every number in this report is generated from result files by `report/make_numbers.py`.